Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/09/2005
PI Name:	Saether, Erik Ph.D.		
Project Title:	Damage-tolerant High-performance Radiation Protectant Sandwich Structural Materials		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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City:	Hampton	**State:**	VA
Zip Code:	23681-2199	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-UNSOL
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
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Task Description:	We will consider a variety of composite sandwich configurations that use materials with high radiation shielding characteristics (high hydrogen content) as the core. These materials typically have only low load-bearing capability and would otherwise be considered only for non-structural applications but may now be utilized as structural materials through appropriate choice of configurational details and judicious use of supplementary high strength structural materials. These composite sandwich configurations will be analyzed to determine their suitability for structural load carrying ability, damage tolerance and radiation shielding potential. Promising configurations will be manufactured and tested under radiation shielding and a variety of structural conditions. These configurations will be ranked according to their radiation shielding and structural performance. The current paradigm for spacecraft design involves the use of an aluminum shell to provide structural integrity combined with heavy parasitic shielding to supplement the radiation shielding requirements of the craft. These designs tend to be heavy and therefore are costly to transport to orbit because the shielding material adds significant weight but does not carry load or provide damage tolerance to the spacecraft. The objective of this task is to develop novel structural configurations to reduce weight of spacecraft while maintaining radiation shielding, and structural and damage tolerance requirements. This project will make clear contributions towards the goal of developing efficient structures for space applications. The fabrication of damage tolerant, radiation protectant structures addresses the need for new, advanced material applications offering weight reduction for near-term space structures, radiation shielding of personnel involved in future extended space missions, and addresses the issue of damage tolerance which relates to structural integrity and ageing spacecraft considerations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	All activities are currently in accordance with the original WBS with only minor scheduling changes. Specific efforts that have been completed or commenced during the period from Sept. 2004 to Sept. 2005 are listed below. Out-of-plane tensile testing of candidate multi-functional sandwich composites: Testing has been performed using several candidate sandwich configurations to ensure proper manufacturing and to assess structural integrity. Testing of unstitched sandwich configurations is complete. Testing of the same sandwich configurations with through-the-thickness stitching is underway to assess performance improvements. Interlaminar shear testing of candidate multi-functional sandwich composites: Test method development for thick, multifunctional sandwich composites completed. Fabrication of test specimens underway. Testing to be based on core shear test method (ASTM C 273) and used to assess the ability of the sandwich configurations to carry interlaminar shear loading. Development of manufacturing techniques for fabricating angle-stitched multi-functional sandwich composites: All stitched sandwich configurations to date have featured stitches oriented perpendicular to the facesheets. For added structural integrity and to increase the shear stability, the fabrication of thick, multifunctional sandwich composites with angled stitches is underway. These angle-stitched sandwich composites, when manufactured, will be subjected to both out-of-plane tensile and interlaminar shear testing. Incorporation of self-healing membrane in multi-functional sandwich composites: A manufacturing method has been developed to incorporate a 3 mm thick layer of Surlyn polymer into a multi-functional stitched sandwich composite. Surlyn has been shown to provide self-healing properties during high velocity impacts. The Surlyn membrane was melt-bonded to adjacent layers of PET foam, inserted into the sandwich configuration, and subsequently stitched through-the-thickness. Sandwich specimens with self-healing membranes will be subjected to both out-of-plane tensile and interlaminar shear testing. Determination of chemical composition of multi-functional sandwich composites and their constituents: Based on an areal weight of 5 gm/cm^2, the chemical composition and elemental percentages were determined for a variety of facesheet materials (fibers and resins), foam core materials, and as-fabricated sandwich configurations for use in radiation shielding simulations. Fabrication of thick-facesheet multifunctional sandwich configurations: Two multifunctional sandwich configurations, each with roughly equal areal weights in the facesheets and cores, were designed and fabricated to support radiation shielding simulations. These two sandwich configurations, spectra-epoxy/polyethylene foam and carbon/epoxy polyurethane foam, were fabricated with a total areal weight of 5 gm/cm^2. Following radiation shielding simulations, these sandwich specimens will be submitted for radiation shielding testing. Physical heavy ion radiation testing: Earlier testing of core materials were performed on specimens with an areal density of approximately 2.0 g/cm2 which yielded fragment profiles near the accuracy of the ion fragment detectors and produced experimental data without sufficient resolution. Thus, an additional set of specimens have been prepared with a higher areal density of 5.0 g/cm2. These specimens include PET and Polyproplyene foam blocks, Spectra and Polyurethane/carbon facesheets and PET/Spectra facesheet composite. We are currently awaiting the sequence of ion beams to be used in the main particle accelerator to schedule testing of our samples. Analytical prediction of radiation shielding capability: Two analysis codes developed at NASA Langley are being used to simulate radiation shielding and mission-dependent space environment to determine equivalent radiation dose subjected to vehicle crews. These are referred to as the "Lab" and "Space" code, respectively. A sequence of amorphous polymers and multilayer sandwich cross-sections with an areal weight of 5 gm/cm^2 have been selected for analysis. A first set of analyses that have been completed involve the Lab code to obtain measures of shielding capability by predicting nucleonic fragmentation of heavy ion impact. These data will be included with the set of physical specimens sent to BNL for testing.
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